

MURCIA, Spain

WMO: 084300

Lat: 38.0019N

Lon: 1.1708W

Elev: 203

StdP: 14.59

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	36.8	39.3	19.1	14.5	50.4	23.6	18.0	50.7	21.1	59.1	18.4	58.4	3.0	270	0.368

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.0	97.9	71.7	95.2	71.2	92.8	71.0	76.8	89.2	75.5	87.0	74.4	85.4	8.5	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.6	126.1	80.8	72.3	120.5	80.3	71.1	115.5	79.9	40.2	89.0	39.0	87.3	37.9	85.5	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.2	14.8	13.1	31.3	104.0	2.5	4.3	29.5	107.1	28.1	109.7	26.7	112.1	24.8	115.3		
(5)			28.6	79.5	3.0	2.4	26.5	81.2	24.7	82.6	23.1	84.0	20.9	85.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	66.6	52.9	54.6	58.6	63.0	69.5	77.2	82.2	82.3	75.9	69.0	59.1	53.7		
	DBStd	11.50	4.87	4.96	5.08	4.60	4.61	4.37	3.29	3.23	4.28	4.65	5.39	4.68		
	HDD50	63	24	15	2	0	0	0	0	0	0	0	3	19		
	HDD65	1533	376	292	207	90	13	0	0	0	0	17	188	350		
	CDD50	6118	113	144	269	391	605	817	999	1002	778	590	276	134		
	CDD65	2113	0	1	9	31	153	367	534	537	328	142	11	0		
	CDH74	20892	3	13	134	304	1248	3618	6028	5961	2708	810	63	2		
	CDH80	9346	0	1	23	45	348	1631	3131	2964	1012	186	5	0		
(14) Wind	WSAvg	5.2	4.7	5.2	5.6	5.7	5.8	6.1	6.1	5.6	4.7	4.1	4.5	4.1		
(15) Precipitation	PrecAvg	11.1	0.6	1.0	1.2	1.3	1.3	0.9	0.1	0.3	0.9	1.6	1.3	0.9		
	PrecMax	22.6	2.3	4.2	5.7	3.9	5.1	3.5	0.4	1.2	5.2	6.9	5.2	3.0		
	PrecMin	4.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	4.5	0.6	1.1	1.2	1.0	1.0	0.9	0.1	0.4	1.2	1.9	1.3	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.1	75.9	82.9	84.6	91.6	99.2	102.5	101.7	95.5	88.4	80.3	72.7		
		MCWB	56.9	56.6	59.8	62.4	66.1	69.8	73.0	73.6	70.2	67.0	61.8	58.1		
	2%	DB	68.9	71.7	77.6	80.4	86.9	94.8	98.0	97.6	91.6	84.5	75.4	68.6		
		MCWB	54.9	55.5	58.5	60.6	64.5	69.1	71.8	72.6	70.4	66.7	60.3	56.4		
	5%	DB	65.6	68.4	73.7	77.4	83.7	91.5	95.2	94.8	88.6	81.7	71.9	65.6		
		MCWB	53.0	54.4	56.9	59.2	63.5	68.6	71.5	72.2	69.8	66.2	58.7	54.9		
	10%	DB	62.3	65.0	70.3	74.4	80.8	88.6	92.7	92.3	85.7	78.6	68.8	62.6		
		MCWB	51.5	52.7	55.3	58.0	62.7	67.6	71.2	72.1	69.2	65.0	57.5	53.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.5	60.0	62.6	65.2	69.9	75.8	78.7	78.7	76.6	71.2	64.9	61.0		
		MCDB	70.6	70.7	77.6	79.5	85.1	90.4	93.6	93.4	86.8	80.7	73.5	68.4		
	2%	WB	56.1	57.2	60.0	62.8	67.3	72.8	76.5	77.1	74.2	69.5	62.9	58.3		
		MCDB	66.1	67.8	73.8	76.5	80.9	86.6	89.7	89.2	84.2	78.9	70.6	65.2		
	5%	WB	54.3	55.4	58.2	61.0	65.6	71.1	75.1	75.8	72.9	68.3	61.1	56.4		
		MCDB	62.6	65.4	70.3	73.0	78.1	84.4	87.8	86.9	82.6	77.1	68.3	63.3		
	10%	WB	52.7	53.9	56.6	59.4	64.2	69.7	73.9	74.8	71.6	67.0	59.5	54.6		
		MCDB	60.3	62.5	67.1	70.6	76.2	83.0	85.7	85.3	80.8	75.2	66.5	61.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.6	19.4	20.7	20.4	20.1	20.8	20.7	20.0	19.0	18.3	17.8	17.7		
		MCDBR	23.3	25.0	27.2	25.5	24.7	25.0	25.0	24.1	23.1	22.5	22.1	21.6		
		MCWBR	12.3	12.6	12.2	10.4	8.7	7.9	7.6	8.0	8.1	8.7	10.7	11.7		
		MCDBR	20.0	21.9	23.8	22.3	21.1	21.4	21.2	20.5	18.9	18.7	17.9	18.7		
	5% WB	MCWBR	11.6	12.1	11.6	9.9	8.4	7.7	7.4	7.5	7.2	7.8	9.3	11.0		
(40) Clear-Sky Solar Irradiance	taub		0.314	0.341	0.391	0.404	0.432	0.455	0.461	0.465	0.433	0.396	0.347	0.322		
	taud		2.462	2.403	2.258	2.253	2.201	2.159	2.153	2.146	2.239	2.335	2.438	2.462		
	Ebn at Noon		275	280	275	278	271	264	261	257	258	257	260	263		
	Edh at Noon		26	31	40	43	46	48	48	47	40	33	27	24		
(44) All-Sky Solar Radiation	RadAvg		810	1073	1453	1795	2119	2393	2374	2064	1607	1184	848	705		
	RadStd		78	110	121	128	149	77	55	73	86	92	78	57		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.65	N/A	N/A	+1.67	+1.42	+1.04	N/A	N/A	+333	+238
Regional (5 neighbors)	N/A	N/A	-2.34	+0.92	N/A	N/A	N/A	N/A	N/A	+104

Nomenclature: See separate page